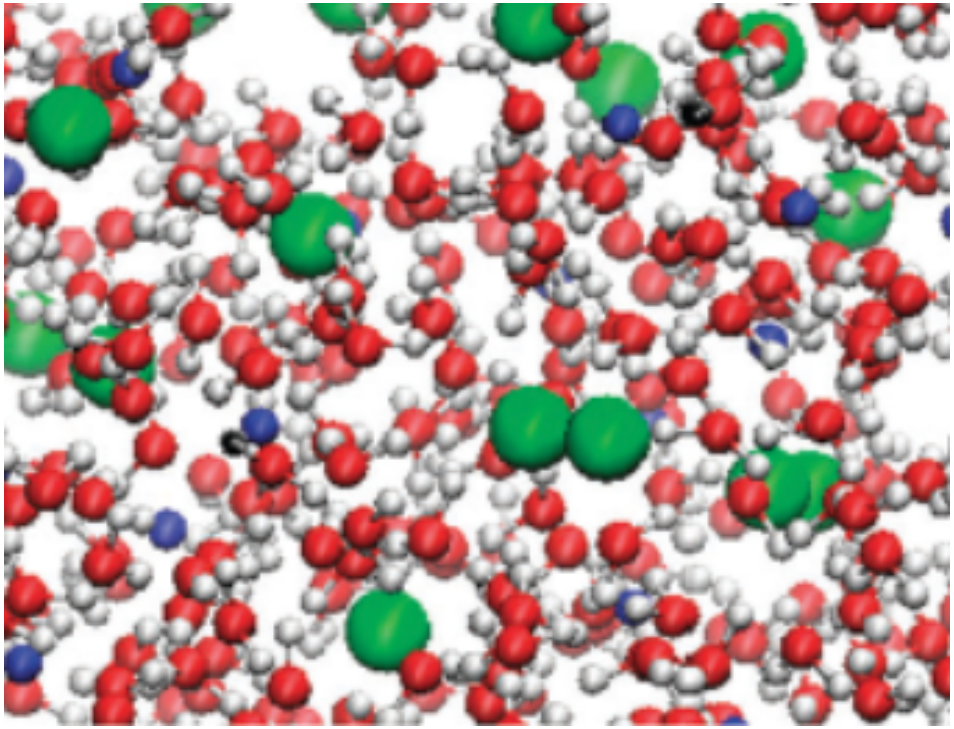
Project: Ionic solution in water



- **Tools:** NAMD (MD simulation), VMD (visualization and analysis), CHARMM-GUI (generation of setups). If you need to install NAMD and VMD, go to University of Illinois-Urbana Champaign: https://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=NAMD
- https://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=VMD

- **Objectives:**

- Create a realistic model of an aqueous solution, composed by MgCl2, in an aqueous 2 M solution at 303 K, then run a full MD simulation (equilibration + production) and analyze microscopical structure of several atomic species.
- Analyze the structure, hydrogen-bonding and single water dynamics using VMD tools. Show snapshots of relevant cases
- Study the case of different temperatures
- Change the concentration of ions and compare with previous results
- Change the ion pair and check the results compared with previous ones

- STEP 1: Preparing Setup. Once VMD has been installed, visualize your system, generated previously (download the set of files 'charmm-gui-MgCl2-water.tar' and unzip it.

- Use the instructions:

- 1. 'vmd ';
- 2. File -> New molecule -> Browse: Go to -> charmm-gui-NaCl-water/namd and select 'step3_pbcsetup.psf' (+ 'Open') -> Load; and then do
- 3. Browse: 'step3_pbcsetup.pdb (+ 'Open') -> Load...and have a look.
- HINT1: Try to use the tools "Graphics -> Representations -> etc. Can use the VMD user guide: https://www.ks.uiuc.edu/Research/vmd/current/ug/
- HINT2: it is clearer if using "Display" -> Orthographic

- STEP 2: Equilibrating the system.

- Series of jobs:
- 1. Go to -> charmm-gui-MgCl2-water/namd
- 2. Copy the engine NAMD file here (namd2)
- 3. Make "./namd2 step4_equilibration.inp"
- This will run 125000 steps' energy minimization + MD, it may take 1 hour, depending on architecture and processor of your laptop. After this run, the equilibration should be ready.

- STEP 3: Production run + analysis of the data collected.

- Now you should make the production run: "./namd2 step5_production.inp > output.log &" (500000 steps, total length 1 ns, if time step is set to 2 fs). Several hours. Estimate how long will it take before running it.
- After the long run ends, have a look at the final movie. To visualize it, use VMD: /home/…/charmm-gui/vmd/ step3_pbcsetup.psf ./namd/step5_production.dcd &
- Check the energy, temperature and pressure versus time. With the "log" file "output.log", you can use the tool "NAMD plot". You should get energy and temperature fluctuating around given mean values. What happens to pressure? Explain the results.

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- STRUCTURE: Radial distribution functions of Ow-Hw, Ow-Na, Hw-Cl, Ow-K: Use 'Extensions' -> 'Analysis' -> 'Radial Pair Distribution Function g(r)'. In 'Selection 1' and '2', write: 'type OT' and 'type HT' for g(r)_{OH}. The numbers of frames run from 0 to N-1 (according the count shown at the main screen). Use 'type CLA' for chloride and 'type MG' for magnesium. Obtain the corresponding coordination numbers n(r) to get complementary information: use 'Display Int(g(r))' from the same 'Radial' tool. Explain your results in physical terms, using the atomic electrical forces as "leitmotif".
- Hydrogen bonding: Using the tool Extensions' -> 'Analysis' -> 'Hydrogen Bonds', calculate the number of HB per water molecule in the system. Set 'Selection 1' as 'resname TIP3' and set HB definition to 'Donor-Acceptor distance = 3.6 Å' and 'Angle cutoff = 30'. Knowing that bulk water has about 4 HBs per molecule, explain your results.

Questions:

- Based on what you learned up to now, explore the influence of temperature on microscopic properties of water and ions. Hint: Try equilibration + "short" production runs (about 30000 steps) at temperatures below 273 and above 373 K.

- What is the effect of the ion concentration on your simulations? Try c = 3, 5 M. To do this, you should use the tool CHARMM-GUI (https://www.charmm-gui.org/?doc=input/solution)

- How could you make the simulations more realistic? What particles could be included? Try a case (for instance N = 100, T = 303 K) including such particles. To do this, prepare a new setup.